An Analysis of Dropout regularization

Dropout regularization

$$\tilde{h} = r \odot h, \quad r_i \sim \text{Bernoulli}(p)$$

Dropout regularization -- Part 1

Early stopping can be viewed as regularization in time. Intuitively, a training procedure such as gradient descent tends to learn more and more complex functions with increasing iterations. By regularizing for time, model complexity can be controlled, improving generalization. Early stopping is implemented using one data set for training, one statistically independent data set for validation and another for testing. The model is trained until performance on the validation set no longer improves and then applied to the test set.

Dropout regularization -- Part 2

In mathematics, statistics, finance, and computer science, particularly in machine learning and inverse problems, regularization is a process that converts the answer to a problem to a simpler one. It is often used in solving ill-posed problems or to prevent overfitting. Although regularization procedures can be divided in many ways, the following delineation is particularly helpful:

Dropout regularization -- Part 3

Enforcing a sparsity constraint on w {\displaystyle w} can lead to simpler and more interpretable models. This is useful in many real-life applications such as computational biology. An example is developing a simple predictive test for a disease in order to minimize the cost of performing medical tests while maximizing predictive power. A sensible sparsity constraint is the L 0 {\displaystyle L_{0}} norm ■ w ■ 0 {\displaystyle \|w\|_{0}} , defined as the number of non-zero elements in w {\displaystyle w} . Solving a L 0 {\displaystyle L_{0}} regularized learning problem, however, has been demonstrated to be NP-hard. The L 1 {\displaystyle L_{1}} norm (see also Norms) can be used to approximate the optimal L 0 {\displaystyle L_{0}} norm via convex relaxation. It can be shown that the L 1 {\displaystyle L_{1}} norm induces sparsity. In the case of least squares, this problem is known as LASSO in statistics and basis pursuit in signal processing.

Dropout regularization -- Part 4

I S [f n] = 1 n ∑ i = 1 N V (f n (x ^ i) , y ^ i) {\displaystyle I_{S}[f_{n}]={\frac {1}{n}}\sum _{i=1}^{N}V(f_{n}({\hat {x}}_{i}),{\hat {y}}_{i})}

Dropout regularization -- Part 5

Other uses of regularization in statistics and machine learning Bayesian learning methods make use of a prior probability that (usually) gives lower probability to more complex models. Well-known model selection techniques include the Akaike information criterion (AIC), minimum description length (MDL), and the Bayesian information criterion (BIC). Alternative methods of controlling overfitting not involving regularization include cross-validation. Examples of applications of different methods of regularization to the linear model are:

Dropout regularization -- Part 6

min w ∈ R p 1 n ■ X ^ w − Y ^ ■ 2 + λ (α ■ w ■ 1 + (1 − α) ■ w ■ 2 2) , α ∈ [0 , 1] {\displaystyle \min _{w\in \mathbb {R} ^{p}}{\frac {1}{n}}\left\|{\hat {X}}w-{\hat {Y}}\right\|^{2}+\lambda \left(\alpha \left\|w\right\|_{1}+(1-\alpha)\left\|w\right\|_{2}^{2}\right),\alpha \in [0,1]}

Dropout regularization -- Part 7

R (w) = ∑ g = 1 G ■ w g ■ 2 , {\displaystyle R(w)=\sum _{g=1}^{G}\left\|w_{g}\right\|_{2},} where ■ w g ■ 2 = ∑ j = 1 | G g | (w g j) 2 {\displaystyle \|w_{g}\|_{2}={\sqrt {\sum _{j=1}^{|G_{g}|}\left(w_{g}^{j}\right)^{2}}}}

Dropout regularization -- Part 8

w T = γ n ∑ i = 0 T − 1 (I − γ n X ^ T X ^) i X ^ T Y ^ {\displaystyle w_{T}={\frac {\gamma }{n}}\sum _{i=0}^{T-1}\left(I-{\frac {\gamma }{n}}{\hat {X}}^{\mathsf {T}}{\hat {X}}\right)^{i}{\hat {X}}^{\mathsf {T}}{\hat {Y}}}

Dropout regularization -- Part 9

Explicit regularization is regularization whenever one explicitly adds a term to the optimization problem. These terms could be priors, penalties, or constraints. Explicit regularization is commonly employed with ill-posed optimization problems. The regularization term, or penalty, imposes a cost on the optimization function to make the optimal solution unique. Implicit regularization is all other forms of regularization. This includes, for example, early stopping, using a robust loss function, and discarding outliers. Implicit regularization is essentially ubiquitous in modern machine learning approaches, including stochastic gradient descent for training deep neural networks, and ensemble methods (such as random forests and gradient boosted trees). In explicit regularization, independent of the problem or model, there is always a data term, that corresponds to a likelihood of the measurement, and a regularization term that corresponds to a prior. By combining both using Bayesian statistics, one can compute a posterior, that includes both information sources and therefore stabilizes the estimation process. By trading off both objectives, one chooses to be more aligned to the data or to enforce regularization (to prevent overfitting). There is a whole research branch dealing with all possible regularizations. In practice, one usually tries a specific regularization and then figures out the probability density that corresponds to that regularization to justify the choice. It can also be physically motivated by common sense or intuition. In machine learning, the data term corresponds to the training data and the regularization is either the choice of the model or modifications to the algorithm. It is always intended to reduce the generalization error, i.e. the error score with the trained model on the evaluation set (testing data) and not the training data. One of the earliest uses of regularization is Tikhonov regularization (ridge regression), related to the method of least squares.

Dropout regularization -- Part 10

This regularizer constrains the functions learned for each task to be similar to the overall average of the functions across all tasks. This is useful for expressing prior information that each task is expected to share with each other task. An example is predicting blood iron levels measured at different times of the day, where each task represents an individual.

Dropout regularization -- Part 11

Dropout In the context of neural networks, the Dropout technique repeatedly ignores random subsets of neurons during training, which simulates the training of multiple neural network architectures at once to improve generalization.

Dropout regularization -- Part 12

Classification Empirical learning of classifiers (from a finite data set) is always an underdetermined problem, because it attempts to infer a function of any x {\displaystyle x} given only examples x 1 , x 2 , … , x n {\displaystyle x_{1},x_{2},\dots ,x_{n}} . A regularization term (or regularizer) R (f) {\displaystyle R(f)} is added to a loss function:

Dropout regularization -- Part 13

w T = (I − γ n X ^ T X ^) γ n ∑ i = 0 T − 2 (I − γ n X ^ T X ^) i X ^ T Y ^ + γ n X ^ T Y ^ = γ n ∑ i = 1 T − 1 (I − γ n X ^ T X ^) i X ^ T Y ^ + γ n X ^ T Y ^ = γ n ∑ i = 0 T − 1 (I − γ n X ^ T X ^) i X ^ T Y ^ {\displaystyle {\begin{aligned}w_{T}&=\left(I-{\frac {\gamma }{n}}{\hat {X}}^{\mathsf {T}}{\hat {X}}\right){\frac {\gamma }{n}}\sum _{i=0}^{T-2}\left(I-{\frac {\gamma }{n}}{\hat {X}}^{\mathsf {T}}{\hat {X}}\right)^{i}{\hat {X}}^{\mathsf {T}}{\hat {Y}}+{\frac {\gamma }{n}}{\hat {X}}^{\mathsf {T}}{\hat {Y}}\\[1ex]&={\frac {\gamma }{n}}\sum _{i=1}^{T-1}\left(I-{\frac {\gamma }{n}}{\hat {X}}^{\mathsf {T}}{\hat {X}}\right)^{i}{\hat {X}}^{\mathsf {T}}{\hat {Y}}+{\frac {\gamma }{n}}{\hat {X}}^{\mathsf {T}}{\hat {Y}}\\[1ex]&={\frac {\gamma }{n}}\sum _{i=0}^{T-1}\left(I-{\frac {\gamma }{n}}{\hat {X}}^{\mathsf {T}}{\hat {X}}\right)^{i}{\hat {X}}^{\mathsf {T}}{\hat {Y}}\end{aligned}}}